

Machine Learning

Session 6

Agenda:

1	The Curse of Dimensionality
2	Feature selection
3	Dimension reduction

The Curse of Dimensionality

The Curse of Dimensionality

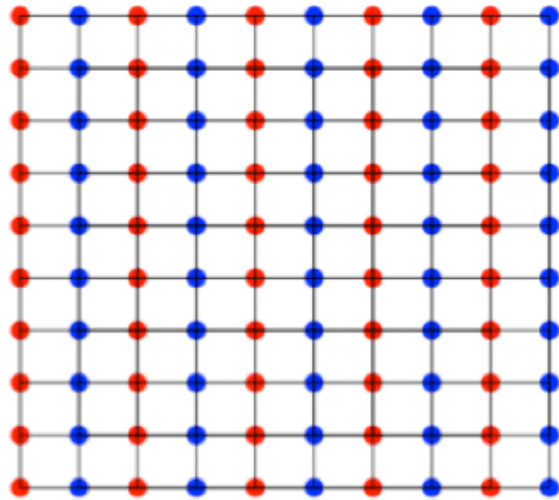
The **Curse of Dimensionality** refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces, often occurring when dealing with datasets that have a large number of features. As the dimensionality of the data increases, several problems arise, making it harder for machine learning algorithms to train efficiently and effectively.



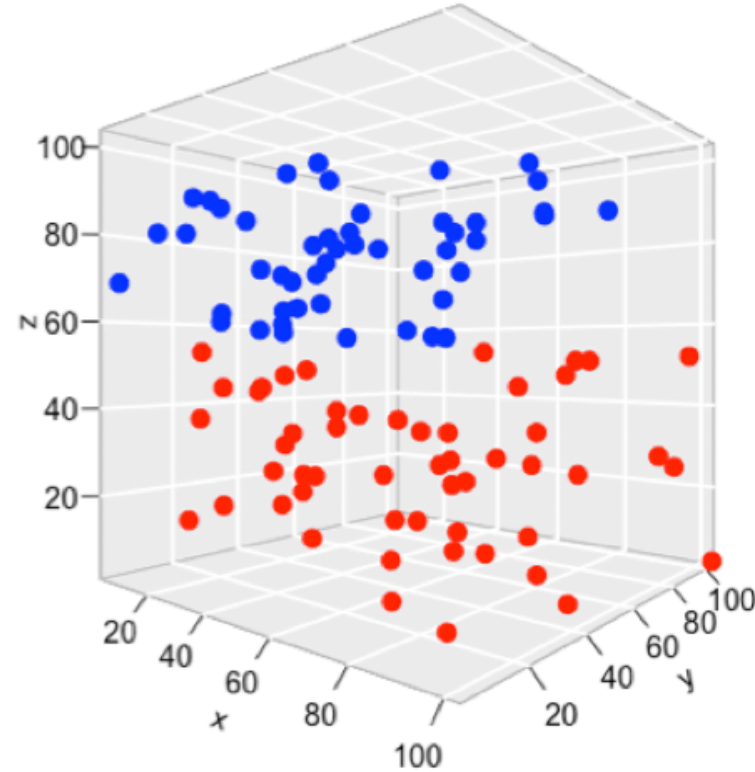
The Curse of Dimensionality



(A) 1-D



(B) 2-D



(C) 3-D

The Curse of Dimensionality

Key Problems Caused by High Dimensionality

1. Increased Computational Requirements

- As the number of features (dimensions) increases, the number of required computations increases exponentially, leading to more computational resources and time.



The Curse of Dimensionality

Key Problems Caused by High Dimensionality

2. Sparsity of Data

- In high-dimensional spaces, the available data becomes sparse, meaning that data points become more spread out. This makes it difficult for algorithms, particularly those that rely on distance measures (like KNN), to make accurate predictions.

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Key Problems Caused by High Dimensionality

2. Sparsity of Data

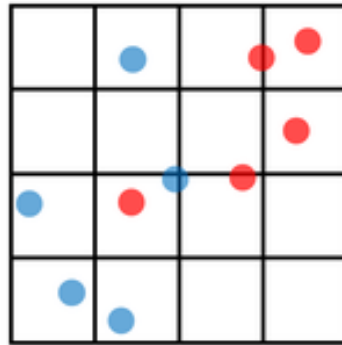
a

1-D Space



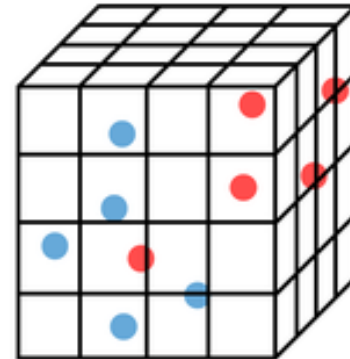
Density: $10/4=2.5$

2-D space



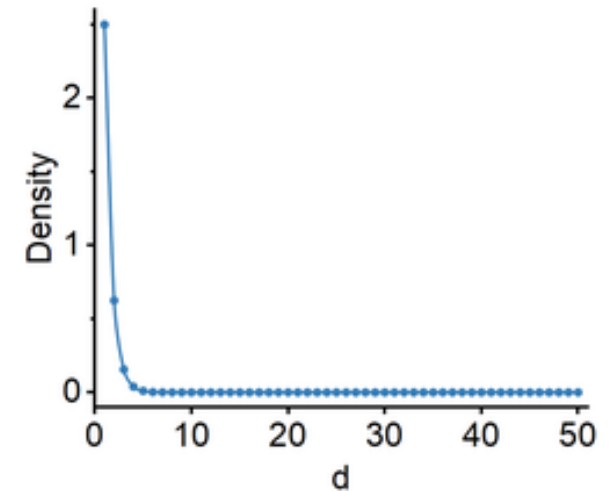
Density: $10/16=0.625$

3-D space



Density: $10/64=0.156$

b



The Curse of Dimensionality

Key Problems Caused by High Dimensionality

3. Overfitting

- With a high number of dimensions, models tend to overfit the training data. The model might rely heavily on noise or irrelevant features, reducing its ability to generalize to unseen data.

The Curse of Dimensionality

Key Problems Caused by High Dimensionality

3. Overfitting



The Curse of Dimensionality

Key Problems Caused by High Dimensionality

4. Redundant and Irrelevant Features

- Many features may be redundant (highly correlated with other features) or irrelevant (do not contribute much to the outcome). This can confuse the model, leading to poorer performance and longer training times.



The Curse of Dimensionality

Key Problems Caused by High Dimensionality

5. Difficult Visualization

- Humans can interpret data well in low dimensions (like 2D or 3D), but as the number of dimensions increases, visualization becomes impossible. We lose the ability to see patterns clearly.

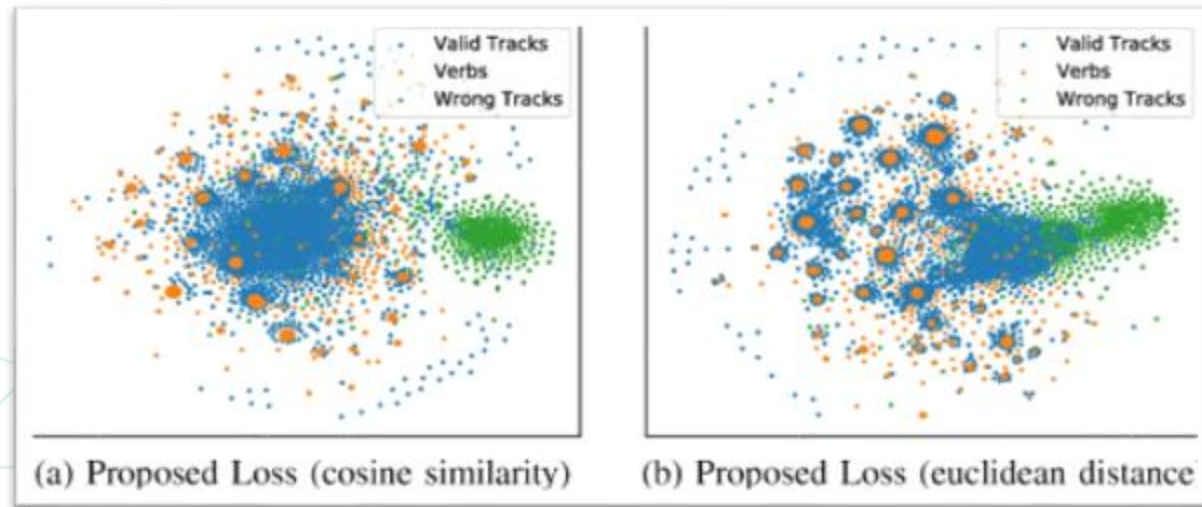


The Curse of Dimensionality

Key Problems Caused by High Dimensionality

6. Distance and Similarity Issues

- Algorithms that rely on measuring distances between data points (e.g., KNN, SVM) become less effective because, in high-dimensional spaces, distances between points tend to converge. This reduces the usefulness of distance-based learning algorithms.



The Curse of Dimensionality

Key Problems Caused by High Dimensionality

7. Storage and Memory Concerns

- High-dimensional datasets require much more storage, making it harder to handle on limited memory systems.



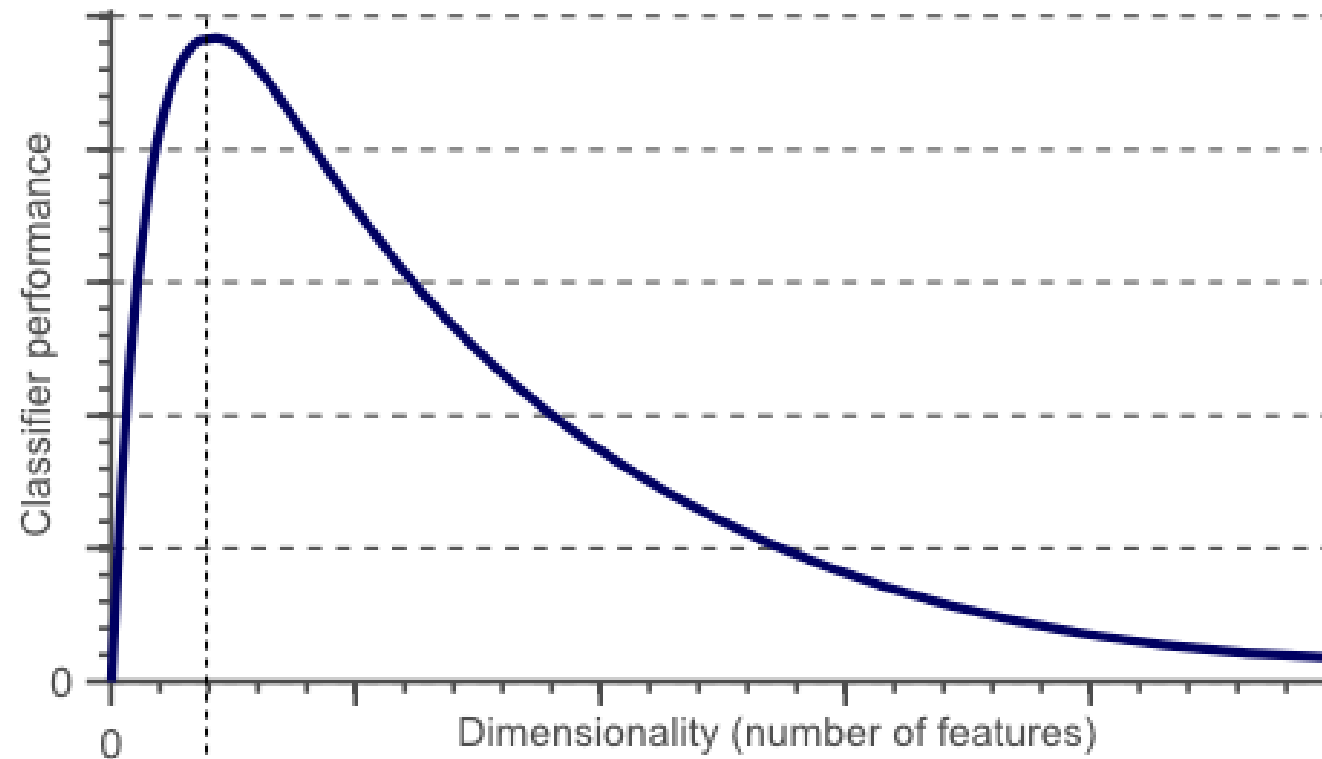
The Curse of Dimensionality

How Dimensionality Reduction Helps ?

Dimensionality Reduction techniques address the curse of dimensionality by reducing the number of features (or dimensions) while retaining as much relevant information as possible. This process helps simplify models, improves computational efficiency, and can even lead to better generalization by reducing overfitting.

The Curse of Dimensionality

How Dimensionality Reduction Helps ?



Feature Selection

The Curse of Dimensionality

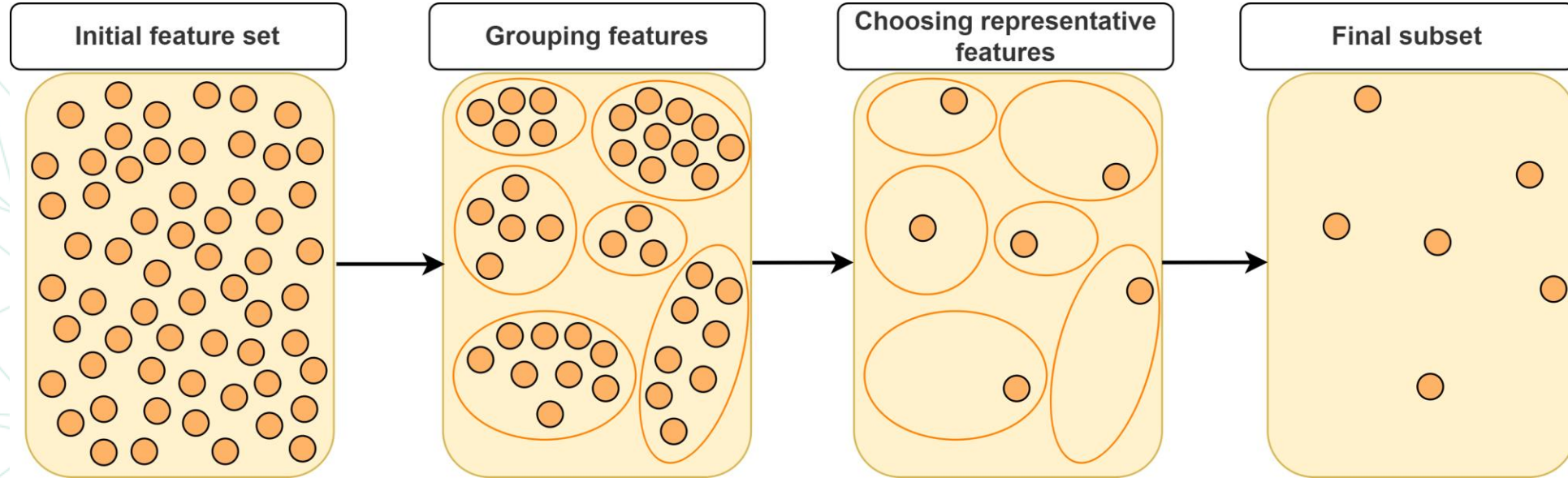
Categories of Dimensionality Reduction

1. Feature Selection

- **Purpose:** This technique involves selecting the most relevant features from the original data and discarding irrelevant or redundant ones.
- **Key Point:** No transformations are performed on the data; instead, we work with a subset of the original features.

The Curse of Dimensionality

1. Feature Selection



The Curse of Dimensionality

Categories of Dimensionality Reduction

1. Feature Selection

Techniques

- **Wrapper Methods:** These methods evaluate feature subsets based on model performance (e.g., Forward and Backward Selection).
- **Filter Methods:** These methods select features based on their statistical relationship with the target (e.g., correlation, mutual information).
- **Embedded Methods:** Feature selection is integrated within model training (e.g., using decision trees, Lasso, or Ridge regression).

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Categories of Dimensionality Reduction

2. Feature Extraction/Projection Techniques (Dimensionality Reduction)

- **Purpose:** This method creates new features by combining the original features, often transforming them into a smaller set of more meaningful variables.
- **Key Point:** Unlike feature selection, feature extraction transforms the features, meaning the final set of features is different from the original input features.

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Categories of Dimensionality Reduction

2. Feature Extraction/Projection Techniques (Dimensionality Reduction)

Techniques

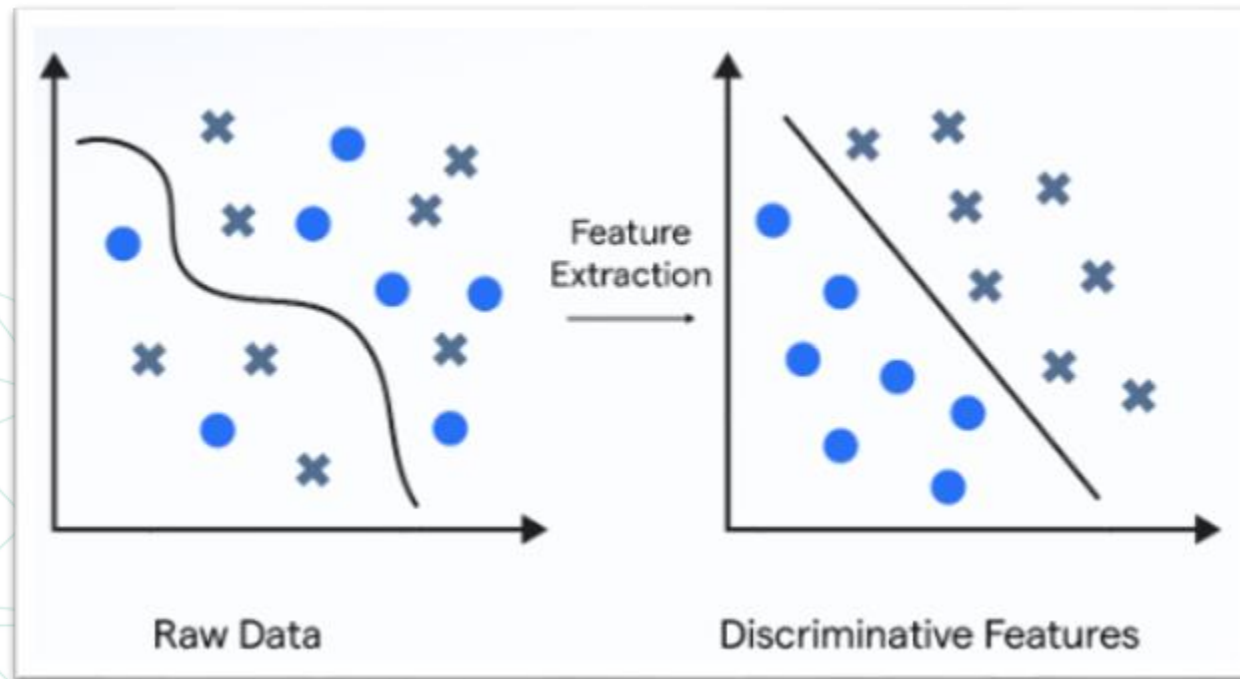
Principal Component Analysis (PCA): A linear transformation technique that reduces dimensionality by projecting data into new axes that capture the most variance.

Linear Discriminant Analysis (LDA): Supervised dimensionality reduction used to project data in a way that maximizes class separation.

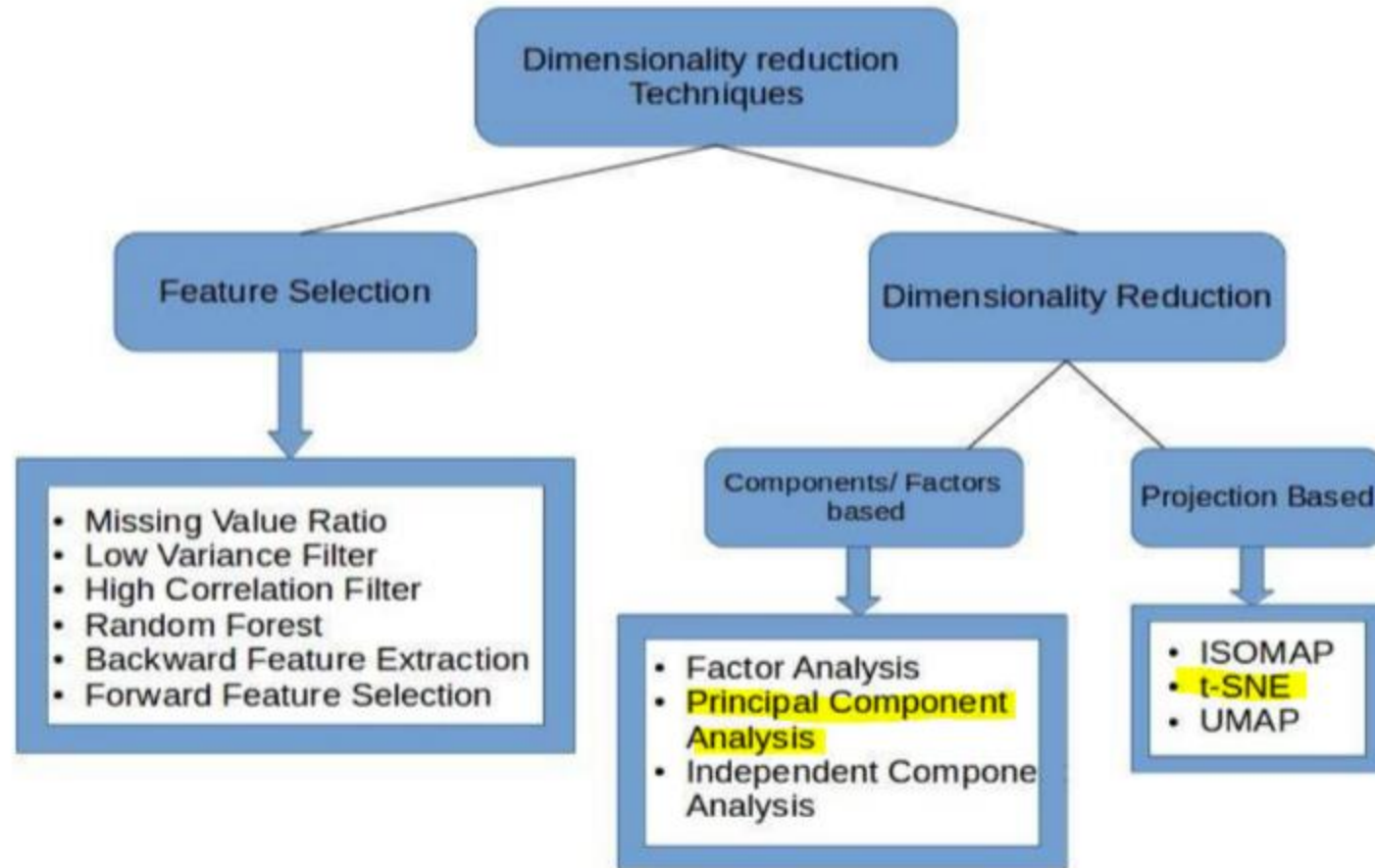
t-SNE and UMAP: Non-linear techniques often used for visualizing high-dimensional data by projecting it into 2D or 3D space.

The Curse of Dimensionality

2. Feature Extraction/Projection Techniques (Dimensionality Reduction)



The Curse of Dimensionality



The Curse of Dimensionality

Feature Selection

1. Filter Methods

- Filter methods select a subset of features based on their relationship with the target variable using **statistical techniques**. These methods are applied as a preprocessing step, independent of any machine learning algorithms.
- **Key Point:** These methods **do not rely on any machine learning models**. Instead, they evaluate the importance of features using statistical tests or correlation with the target variable

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Feature Selection

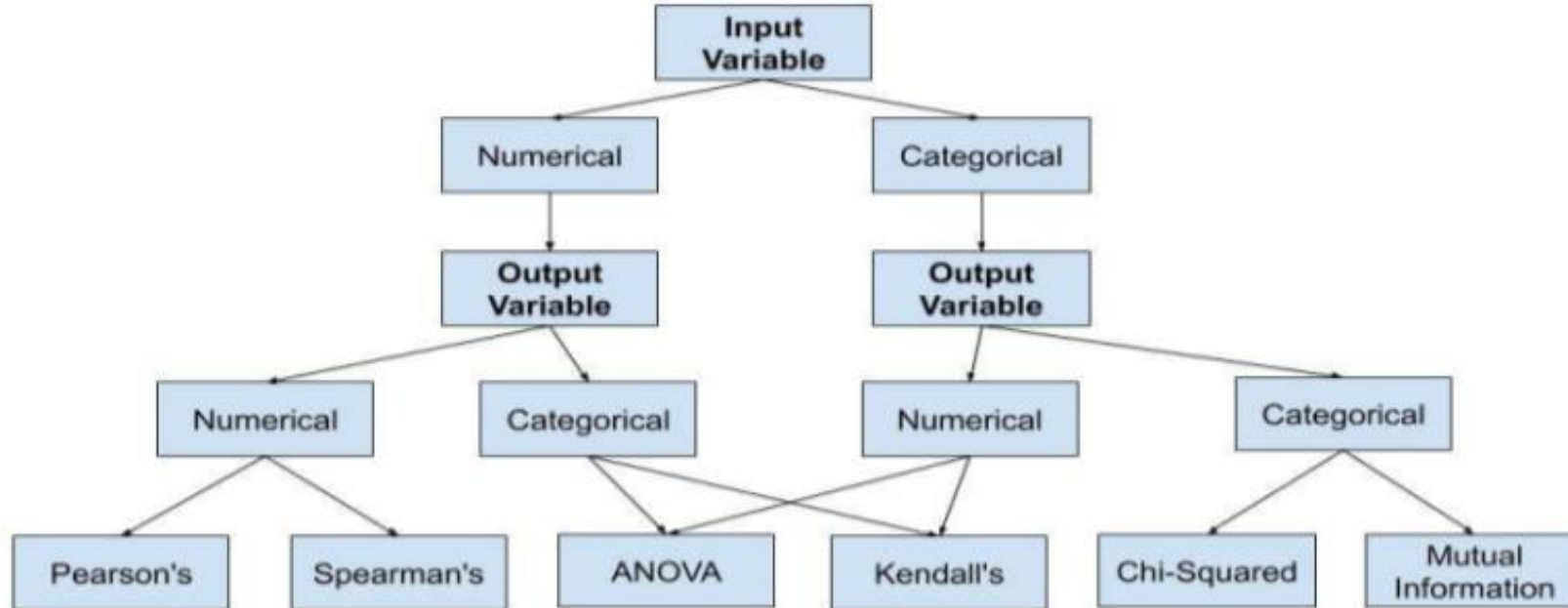
1. Filter Methods

Common Techniques

- **Pearson's correlation:** Measures the linear relationship between features and the target.
- **Chi-square test:** Measures the dependence between two categorical variables.
- **ANOVA (Analysis of Variance):** Measures the difference in **the means of a feature between groups**.
- **Kendall's and Spearman's rank correlation:** Measures ordinal association between variables.

The Curse of Dimensionality

How to Choose a Feature Selection Method



The Curse of Dimensionality

Pearson's Correlation

Pearson's correlation measures the **linear relationship** between two continuous variables. It quantifies how changes in one variable are linearly associated with changes in another.

- Pearson's correlation coefficient (denoted as **r**) ranges from **-1 to +1** indicates a perfect positive linear relationship.
- **-1** indicates a perfect negative linear relationship.
- **0** means there is **no linear relationship** between the variables.

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Pearson's Correlation

- The goal is to find how strongly a feature (independent variable) is linearly correlated with the target variable (dependent variable).
- Variables that have a correlation value above a certain threshold (e.g., 0.5 or below -0.5) are considered strongly correlated with the target, while others are not.

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Chi-Square (χ^2) Test

The Chi-Square test is used to determine if two categorical variables are independent of each other. It's a statistical method to evaluate the association between features and the target variable when both are categorical. The Chi-Square statistic will be low if the variables are independent, and high if there is a strong association between the variables.

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Chi-Square (χ^2) Test

Formula
$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

- O_i = Observed value
- E_i = Expected value
- The sum is calculated over all categories.

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Chi-Square (χ^2) Test

Steps

1. Define the Hypothesis

- Null Hypothesis (H_0): The two variables are independent (no relationship).
- Alternative Hypothesis (H_1): The two variables are dependent (have a relationship).

2. Build the Contingency Table

- This is a table of observed frequencies (counts) of the categorical variables. It's often represented as a **confusion matrix**.

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Chi-Square (χ^2) Test

Steps

3. Find Expected Values

- Expected values are calculated based on the assumption that the two variables are independent.

4. Calculate the Chi-Square Value

- Use the formula to calculate the Chi-Square statistic for each combination of categories.

5. Accept or Reject the Hypothesis

- Compare the calculated Chi-Square value to a critical value from the Chi-Square distribution (based on degrees of freedom and significance level, usually 0.05).

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Summary

- **Pearson's correlation** helps measure linear relationships between continuous features and the target.
- **Chi-Square Test** helps assess the dependence between categorical variables.
- **ANOVA** compares the means of features across groups.
- **Spearman's and Kendall's correlation** measure monotonic (rank-based) relationships.

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2. Wrapper Methods

Wrapper methods treat feature selection as a **search problem** where various combinations of features are evaluated, and the model with the best performance is selected. These methods rely on using a machine learning algorithm to evaluate different subsets of features.

- Unlike filter methods, wrapper methods involve **training a model** on different subsets of features and selecting the combination that yields the best performance.

The Curse of Dimensionality

2. Wrapper Methods

Example Techniques

- **Recursive Feature Elimination (RFE):** Recursively removes the least important features based on model performance and retrains the model until the desired number of features is achieved.
- **Forward Selection:** Starts with an empty set of features and adds one feature at a time, selecting the one that improves model performance the most.
- **Backward Elimination:** Starts with all features and removes the least useful ones iteratively until model performance no longer improves.

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3. Embedded Methods

Embedded methods perform feature selection during the model training process itself. The learning algorithm decides which features contribute most to the model's accuracy.

- These methods combine the advantages of both filter and wrapper methods. They are more efficient than wrapper methods and more robust than filter methods because they are integrated with model training.

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3. Embedded Methods

Common Techniques

- **Lasso Regression (L1 Regularization):** Shrinks the coefficients of less important features to zero, effectively removing them from the model.
- **Ridge Regression (L2 Regularization):** Reduces the impact of less important features without fully eliminating them.
- **Random Forest Feature Importance:** Uses tree-based algorithms like Random Forest to compute feature importance based on how much they decrease impurity (e.g., Gini index) in classification tasks.

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Wrapper Methods vs. Embedded Methods

- Both **wrapper methods** and **embedded methods** utilize machine learning algorithms, but they differ in how they approach feature selection.

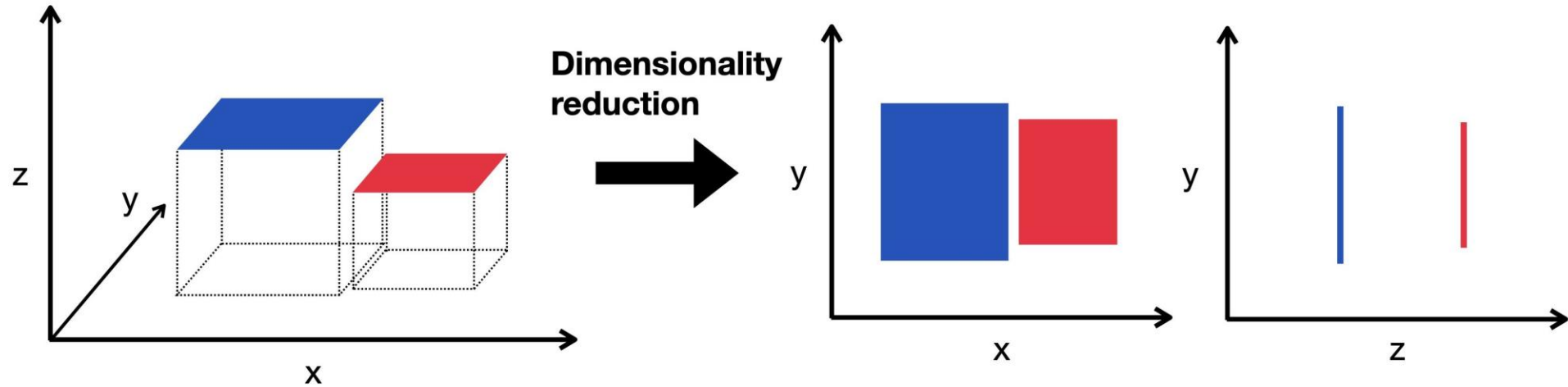
Key Difference

- **Wrapper methods** consider unimportant features **iteratively** and evaluate them based on an external metric, while **embedded methods** handle feature selection **internally** during the model training process.

Dimension reduction

Dimension reduction (Feature extraction)

Feature extraction is a type of dimensionality reduction where transformations take place on the input variables to create new variables (features) that retain most of the information from the original dataset. Instead of removing features, feature extraction combines the input variables in a way that preserves information while reducing the complexity.



Dimension reduction (Feature extraction)

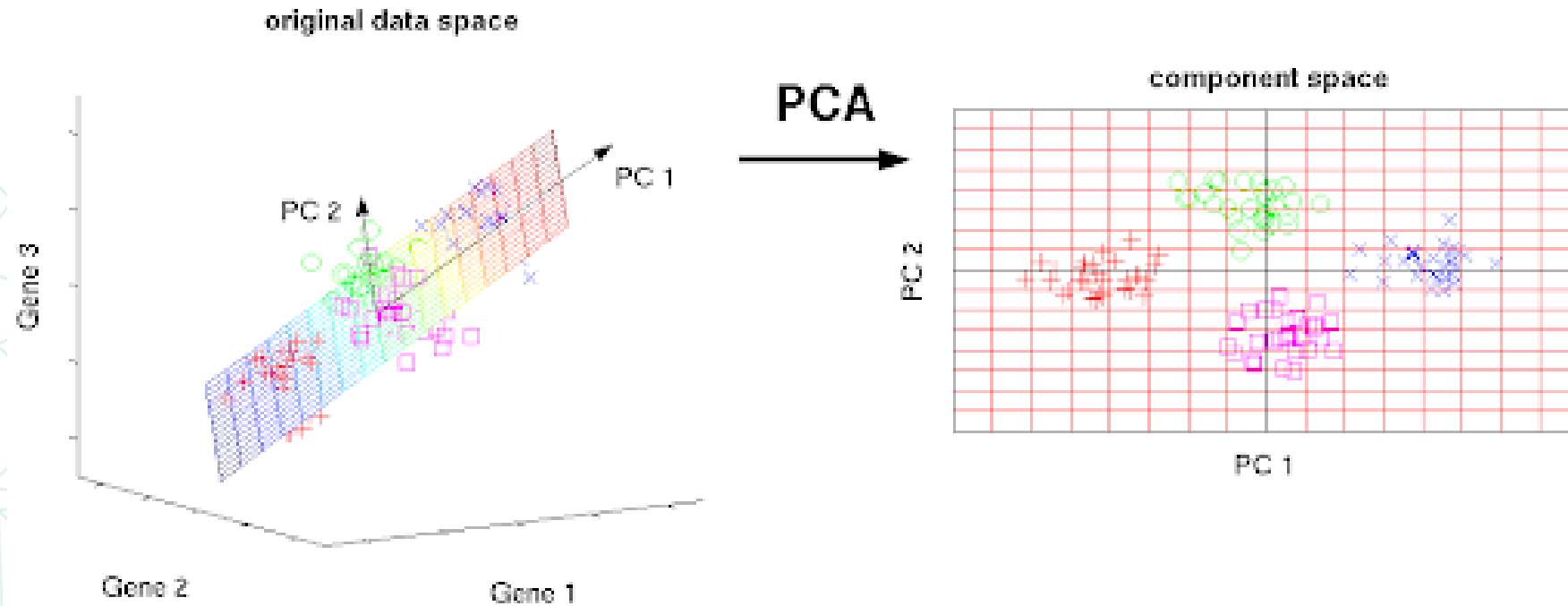
Categories of Feature Extraction

1. Linear Methods (Factor-based)

- **Principal Component Analysis (PCA):** A widely-used linear method that transforms the features into a set of orthogonal (uncorrelated) components. Each component captures the maximum possible variance from the dataset.
- **Factor Analysis (FA):** Similar to PCA, but focuses on modeling the variance of the observed variables by decomposing them into factors.
- **Independent Component Analysis (ICA):** A method that finds components that are statistically independent of each other, commonly used for signal processing.

Dimension reduction (Feature extraction)

Principal Component Analysis (PCA)



Dimension reduction (Feature extraction)

Categories of Feature Extraction

2. Non-Linear Methods (Projection-based or Manifold Learning)

- **Kernel PCA:** An extension of PCA that can handle non-linear relationships between features.
- **t-Distributed Stochastic Neighbor Embedding (t-SNE):** A method used primarily for visualizing high-dimensional data by embedding them into a lower-dimensional space while preserving the structure.
- **Multidimensional Scaling (MDS):** A method that maps the data to a lower-dimensional space while preserving the distances between points.
- **Kohonen Self-Organizing Map (SOM):** An artificial neural network algorithm used for dimensionality reduction and clustering.
- **Uniform Manifold Approximation and Projection (UMAP):** A newer technique that provides faster and more accurate results than t-SNE for high-dimensional data projection.

Dimension reduction (Feature extraction)

Variance and Standard Deviation

Standard Deviation: It indicates how far the values in a dataset are from the mean (average). In simpler terms, it gives an idea of how "spread out" the data points are. It is calculated as the **square root of the variance**.

Formula for standard deviation $\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$

Dimension reduction (Feature extraction)

Variance and Standard Deviation

Variance: It is the average of the squared differences from the mean. Variance emphasizes the spread of the data and helps detect **extreme outliers** by squaring the differences between each data point and the mean.

Formula for variance
$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

Both **variance** and **standard deviation** are used to measure the **spread** of data. Variance is more sensitive to **outliers** because of the squared differences.

Dimension reduction (Feature extraction)

Why Variance Matters in PCA

In **Principal Component Analysis (PCA)**, the goal is to reduce the dimensionality of data while retaining the most **informative features**. PCA identifies the directions (principal components) where the data has the **highest variance**, assuming that directions with higher variance are more informative.

1. **Variance in PCA:** PCA looks for directions with maximum variance to create the principal components. The principal component with the highest variance is considered the most important, as it captures the most information about the spread of the data.
2. **Outliers and Spread:** Variance also helps identify outliers. In PCA, if the spread (variance) is high in a particular direction, that direction is likely to be significant for the dataset's overall structure.

Dimension reduction (Feature extraction)

Linear Transformation

A **linear transformation** is a function that maps a vector space into another vector space while preserving the operations of vector addition and scalar multiplication.

- **Example:** In the image, shapes like squares can be stretched, compressed, or rotated, which are common forms of linear transformations. A key property of a linear transformation is that straight lines remain straight, even after the transformation.
- **PCA (Principal Component Analysis)** uses linear transformations to project high-dimensional data into a lower-dimensional space.

Dimension reduction (Feature extraction)

Matrix Rank

A **linear transformation** is a function that maps a vector space into another vector space while preserving the operations of vector addition and scalar multiplication.

- **Example:** In the image, shapes like squares can be stretched, **compressed**, or **rotated**, which are common forms of linear transformations. A key property of a linear transformation is that straight lines remain straight, even after the transformation.
- **PCA (Principal Component Analysis)** uses linear transformations to project high-dimensional data into a lower-dimensional space.

Dimension reduction (Feature extraction)

Eigenvalues and Eigenvectors

- **Eigenvectors** are special vectors that do not change direction during a transformation. When a matrix is multiplied by one of its eigenvectors, the result is the same eigenvector scaled by a corresponding **eigenvalue**.
- **Eigenvalue**: The factor by which the eigenvector is scaled during the transformation.
- **Example**: In the given example, the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is an eigenvector of the matrix A , and 5 is the corresponding eigenvalue, as shown in the formula $Av=\lambda v$, where v is the eigenvector and λ is the eigenvalue.

Dimension reduction (Feature extraction)

Eigenvalues and Eigenvectors

$$\overset{n \times n}{\text{Matrix}} \vec{A} \vec{x} = \lambda \vec{x}$$

Eigenvector (points to $\vec{x}$)

Eigenvalue (points to λ)

Dimension reduction (Feature extraction)

Eigenvalues and Eigenvectors in PCA

- Eigenvectors in PCA represent the directions (or axes) in which the data is projected.
- Eigenvalues correspond to the magnitude of the variance in those directions. The larger the eigenvalue, the more significant that component is in explaining the variance in the data.

Dimension reduction (Feature extraction)

Pre-conditions to apply PCA

1. Features should be scaled (preferably standardized)

- When performing PCA, it's important to standardize or scale the features. This is because PCA relies on variance, and if the features are not standardized, the features with the largest scale will dominate the principal components due to their larger variance. This would bias the PCA results toward the features with higher variance due to their scale rather than the true structure of the data.

Dimension reduction (Feature extraction)

Pre-conditions to apply PCA

2. PCA applies to numerical data

- PCA is designed for **numerical data**. It doesn't handle categorical data directly. If you have categorical data, it should be transformed into numerical format (e.g., through optimal scaling or encoding) before applying PCA.

Dimension reduction (Feature extraction)

Steps to perform PCA (Algorithm steps)

1. Standardize the data

- Before applying PCA, it is important to standardize the dataset so that each feature has a mean of 0 and a standard deviation of 1. This ensures that all features contribute equally to the analysis.

Dimension reduction (Feature extraction)

Steps to perform PCA (Algorithm steps)

2. Calculate the covariance matrix (Σ)

- After standardizing, compute the **covariance matrix** of the data. The covariance matrix represents the relationships (covariances) between the features. It helps identify the directions of maximum variance in the data.

Dimension reduction (Feature extraction)

Steps to perform PCA (Algorithm steps)

3. Calculate the **eigenvectors** and **eigenvalues** of the covariance matrix

- Compute the **eigenvectors** (which represent the directions of maximum variance) and the corresponding **eigenvalues** (which represent the magnitude of variance in those directions). Sort the eigenvectors in order of their eigenvalues, from largest to smallest.

Dimension reduction (Feature extraction)

Steps to perform PCA (Algorithm steps)

4. Select the first K eigenvectors

- Choose the top **K eigenvectors** that correspond to the largest eigenvalues. These eigenvectors form the **principal components**, which capture the most variance in the data. The number of components (K) is typically chosen to explain the majority of the variance in the dataset.

Dimension reduction (Feature extraction)

Steps to perform PCA (Algorithm steps)

5 . Multiply the original data by the selected eigenvectors

- Project the original data onto the new feature space (principal components) by multiplying the original data matrix by the **K selected eigenvectors**. This transforms the data into a new set of dimensions that are uncorrelated and capture the most variance.

Dimension reduction (Feature extraction)

Principal Component Analysis (PCA) — How to Choose the Number of Components (K)

In PCA, one of the critical steps is to determine how many principal components (K) to retain for reducing the dimensionality of the data while maintaining most of the variance. Here's how to approach it



Dimension reduction (Feature extraction)

What should be the number of components K that we choose?

1. Explained Variance

- The goal is to retain enough components that explain a large proportion of the variance in the data (e.g., 95%). The explained variance is the amount of information (or variance) that the components carry about the data.
- **Tip:** If you are not reducing dimensionality for visualization but for data analysis, retaining components that explain around 95% of the variance is generally a good rule of thumb.

Dimension reduction (Feature extraction)

What should be the number of components K that we choose?

2. Reducing for Visualization

- If you are reducing dimensionality specifically for **data visualization**, it is practical to reduce the number of components down to **2 or 3**. This is because most visualizations (scatter plots, etc.) work in 2D or 3D spaces.

Dimension reduction (Feature extraction)

What should be the number of components K that we choose?

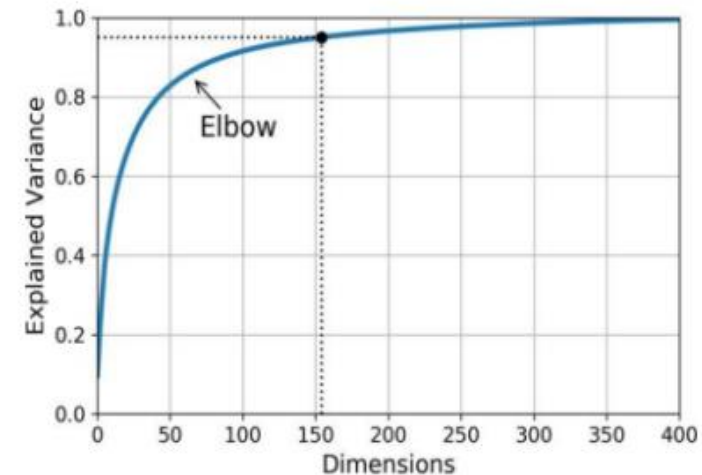
3. Elbow Method

- Another approach is to plot the explained variance as a function of the number of components (K). The elbow point in this curve is where adding more components contributes less to the explained variance.
- The elbow curve suggests a cutoff point where the dimensionality can be reduced without losing much explained variance.

Dimension reduction (Feature extraction)

Example — Elbow Function for PCA

- The elbow method helps visualize the cumulative explained variance as a function of the number of components.
- In the graph example shown, there is an elbow around 100 components, suggesting that reducing the number of components to 100 would still retain much of the original variance while significantly reducing the dimensionality.



Dimension reduction (Feature extraction)

t-SNE Explanation and Steps

1. t-SNE for Dimensionality Reduction

- t-SNE is primarily used for visualization purposes. Unlike PCA, which is linear, t-SNE is a non-linear dimensionality reduction technique that is highly effective in projecting high-dimensional data down to two or three dimensions for visualization.
- While it is not as commonly used in training machine learning models (because it doesn't preserve large-scale structure), t-SNE is valuable for exploratory data analysis.

Dimension reduction (Feature extraction)

t-SNE Explanation and Steps

2. How t-SNE Works

- **High-dimensional Space:** In high dimensions, t-SNE tries to preserve the distance between points. Points that are close to each other in high-dimensional space remain close in lower dimensions. Similarly, points that are far apart remain distant.
- **Attraction and Repulsion:** During the projection process, t-SNE attracts points that are nearby and repels points that are farther away in the original high-dimensional space.
- **Clusters and Visual Interpretation:** After projection, the points form clusters that reveal underlying patterns and groupings in the data.

Dimension reduction (Feature extraction)

Notes

1. Feature Selection before Encoding

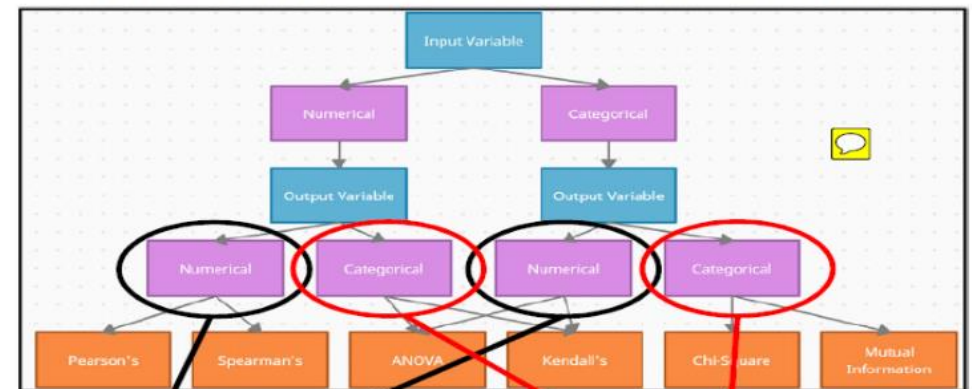
- Feature selection should be applied before the encoding process. The encoding process is applied to convert categorical variables into numerical values.
- After encoding, the data becomes numerical, making it unnecessary to perform feature selection afterward in certain cases.

Dimension reduction (Feature extraction)

Notes

1. Feature Selection before Encoding

Filter-Based Feature Selection Methods



Regression Problem

Pearson's, Spearman's,
ANOVA, Kendall's

Classification Problem

ANOVA, Kendall's,
Chi-Square, Mutual Information

Dimension reduction (Feature extraction)

Notes

2. Handling Text Features

- If the dataset contains text data, then encoding (like one-hot encoding or label encoding) will be needed before any selection process, especially if you're using methods like Chi-square that depend on categorical features.
- For numeric values, encoding is irrelevant, but this step becomes necessary for textual data.

Dimension reduction (Feature extraction)

Notes

3. Order of Operations

- It's important to keep the right order: Feature selection should happen before splitting the data into training and testing sets (train-test split). This ensures that the selected features are consistent across both the training and testing data.
- Scaling (like StandardScaler or MinMaxScaler) should be done after the feature selection process.

Dimension reduction (Feature extraction)

Notes

4. PCA and Scaling

- When using Principal Component Analysis (PCA), scaling is essential. PCA looks for the direction of maximum variance, and without scaling, the features with larger scales may dominate the variance calculation.

Dimension reduction (Feature extraction)

Remarks

1. Chi-square Method and Negative Input

- The Chi-square test does not support negative input values. Hence, if you're working with features that contain negative values, you cannot apply the Chi-square test directly without transforming the data.

2. Feature Name Output

- Some datasets don't include all the feature names, and hence methods like `feature_name_out` may not work unless the dataset is explicitly labeled. This might require additional steps for data preparation or using different tools.